

Business Calculus (MATH 2350)

Summer 2018

Thursday, 05/31/2018

Quiz 3: Section 11.2,11.3

Name (Print): _____

Score: _____ / 100

Section 11.2

1. 15 points Let $f(x) = 9e^x + x^5$. Find the derivative $f'(x)$.

2. 20 points Let $f(x) = \ln(x^6) + 3x^3 + 80$. Find the derivative $f'(x)$.

Hint: $\log_b(x^n) = n\log_b(x)$.

3. 20 points Let $f(x) = 3x + \ln(3x) + e^{3+x}$. Find the derivative $f'(x)$.

Hint: $e^{r+t} = e^r e^t$ and $\log_b(rt) = \log_b(r) + \log_b(t)$.

Section 11.3

1. 15 points Let $f(x) = 3x^4e^x$ Find the derivative $f'(x)$.

2. 15 points Let $f(x) = \frac{1 + e^x}{1 - x}$ Find the derivative $f'(x)$.

3. 15 points Let $f(x) = \frac{x^2 + 2}{2x^2 + x}$ Find the derivative $f'(x)$.

4. 10 points (bonus) Let $f(x) = \frac{(x^3 + 9x^2 + 4)(x^4 + x^3 + x + 1)}{(x^2 + 1)}$. Find the derivative $f'(x)$.